

Repair Request - Database Design

SQL Server Physical Design + Performance Correction

Document Control	Value
Document ID	RR-DBD-001
Version	1.2 - Performance Revision Candidate
Status	Ready for Technical Baseline Approval
Revision Date	12 September 2026
Supersedes	RR-DBD-001 v1.1
Revision Scope	Correct notification index ambiguity; require evidence-based query/index/worker tuning and preserve transaction/concurrency semantics.
Approval	Business Owner / Project Manager + Architecture/QA review - pending

Revision rule / กติกาการแก้ไข: เอกสารฉบับนี้เป็นการปิด Audit ด้าน Performance/Technical Design เท่านั้น ไม่เปลี่ยน State, Role, Guard, SLA, Permission, Notification Recipient หรือ Business Semantics ที่ล็อกไว้เดิม หากคำตัดสินด้าน Performance ใดไม่มีอยู่ใน Business Baseline จะระบุเป็น **Gap / Proposed Technical Decision / Pending Approval** และไม่ถือเป็น Requirement เดิมโดยอัตโนมัติ.

Document control note: หน้านี้เป็น Document Control ที่มีผลสำหรับ revision นี้และ supersede ค่า Version/Status บนหน้าปกของ source baseline ที่ถูกรักษาไว้ในหน้าถัดไปเพื่อ traceability.

Repair Request - Database Design

SQL Server Physical Design / MVP Baseline Revision

Document Control	Value
Document ID	RR-DBD-001
Version	1.1 - Technical Design Baseline Candidate
Status	Ready for Technical Baseline Approval
Revision Date	12 September 2026
Approval	Business Owner / Project Manager - formal name/signature pending
Sources	RR-DD-001 v1.3 + Requirement/BR/State/UC/Test revised baselines
Audience	Database Architect, Backend Developer, SA, QA, Security Reviewer
Language	ภาษาไทยเป็นหลัก พร้อมชื่อ Entity / Role / State / API เป็นภาษาอังกฤษ
Purpose	กำหนด physical SQL Server schema, constraints, transactions, indexes, concurrency and security without changing business semantics.

Revision rule: เอกสารฉบับนี้ปิด Audit findings ของชุดวันที่ 11 September 2026 โดยแยกสิ่งที่เป็น Business Baseline ออกจาก Technical Design อย่างชัดเจน. การลงนามอนุมัติจริงยังเป็นหน้าที่ของ Business Owner / Project Manager.

1. Physical Design Principles

Area	Decision
Primary keys	Transactional/control tables use uniqueidentifier; NEWSEQUENTIALID() may be used for SQL-generated keys.
Concurrency	Logical row_version maps to SQL Server rowversion; every state-changing command checks token + state.
Time	UTC datetime2(3); Site time zone only at presentation.
State	varchar code + CHECK/lookup; transition guards remain service layer; direct state writes prohibited.
Transactions	Convert, Accept/Reject, Corrective creation, Cancel propagation, Close, Time Correction approval and SLA Override are atomic.
Files	Private object storage; DB stores metadata/reference; only CLEAN evidence is usable.
Audit	append-only; business API cannot physical-delete.
Security	Backend enforces tenant/site/role/ownership; guessed IDs outside scope denied without leakage.
Traceability	correlation_id links API, transaction, audit, notification/outbox.

2. Schema Overview

Table / Group	Purpose	Core Key / Constraint
repair_request	Repair Request aggregate	PK repair_request_id; UQ request_no; tenant/site scope; status CHECK; rowversion
approval_route / approval_route_step	Approval routing master	route + ordered steps; UQ(route_id,step_no)
repair_request_approval	Decision history	UQ(request_id,step); decision code incl RETURNED_FOR_CORRECTION
file_asset / request_attachment	Private file metadata and request link	SHA-256; scan status; no public URL
work_order	Execution aggregate	UQ repair_request_id; status; team; acceptance contact ref; rowversion
service_visit	Visit lifecycle	type/status/schedule/assignment; self FK source_missed_visit_id; MISSED decision fields
work_session / time_correction	Technician time	sequence + optimistic concurrency; correction history
work_summary / work_summary_evidence	Versioned result/evidence	UQ(service_visit_id,revision_no); evidence -> file_asset
cost_summary	Close guard	UQ work_order_id; review fields
customer_acceptance_contact	Designated contact	UQ work_order_id; CHECK source_role IN (REQUESTER,SITE_CONTACT)
customer_acceptance	Acceptance rounds	UQ(work_order_id,round_no)
corrective_action	Reject cycles	UQ(work_order_id,cycle_no); rejected acceptance FK
sla_record / sla_override	Resolution SLA	

Table / Group	Purpose	Core Key / Constraint
		UQ repair_request_id; risk_at T-4h; sticky breach; override history
notification_event / notification_recipient	Outbox + recipients	1:N recipient delivery; retry independent
audit_history	Append-only trace	entity_type/id + correlation; no business delete

3. Key Constraint Baseline

```
-- Representative constraints (canonical semantics)
UNIQUE(work_order.repair_request_id)
CHECK(customer_acceptance_contact.source_role IN ('REQUESTER', 'SITE_CONTACT'))
UNIQUE(customer_acceptance_contact.work_order_id)
UNIQUE(customer_acceptance.work_order_id, acceptance_round_no)
UNIQUE(corrective_action.work_order_id, cycle_no)
UNIQUE(sla_record.repair_request_id)
CHECK(file_asset.malware_scan_status IN ('PENDING', 'CLEAN', 'FAILED'))
CHECK(service_visit.status IN ('SCHEDULED', 'RESCHEDULED', 'IN_PROGRESS', 'COMPLETED', 'MISSED', 'CANCELLED'))
-- status changes occur only through allowed service/API commands.
```

4. Core Transaction Boundaries

Transaction	Atomic Steps / Expected Result
Convert Request -> WO	Validate APPROVED + rowversion -> insert unique WO OPEN -> Request CONVERTED -> audit + outbox. Any failure rollback and Request stays APPROVED.
Customer Reject	Insert Acceptance(REJECT) -> create CorrectiveAction(DRAFT) -> WO CORRECTIVE_ACTION_REQUIRED -> audit/outbox.
Customer Accept	Insert Acceptance(ACCEPT) -> WO COMPLETED -> audit/outbox; duplicate decision rejected.
Cancel Work Order	Validate Supervisor/pre-accept -> WO CANCELLED -> cancel non-final visits -> SLA stop -> audit/outbox; COMPLETED/MISSED history remains.
Close Work Order	Validate COMPLETED + reviewed Summary + reviewed Cost -> CLOSED -> SLA stop -> audit.
MISSED decision	Update original MISSED decision fields; optionally insert new SCHEDULED child Visit; original status never changes.
SLA Override	Insert override -> update effective_due_at + risk_at -> audit; preserve original_due/ breached_at.
Notification delivery	Business transaction inserts event+recipients; provider send/retry runs independently after commit.

5. Index / Performance Strategy

Use Case	Index Strategy
Request List/Search	tenant_id + site_id + status + submitted_at; include category/priority/request_no as query requires
Duplicate warning	tenant+site+category+equipment/location+submitted_at with terminal/state filtering
SLA monitor	stopped_at, breached_at, risk_at, effective_due_at

Use Case	Index Strategy
Technician schedule	assigned_technician_id + status + scheduled_start_at
Audit timeline	tenant_id + entity_type + entity_id + occurred_at DESC
Approval inbox	assigned_approver_id + status + routed_at
Notification inbox/retry	recipient_user_id + delivery_status + created_at; failed retry by status/attempt

6. Referential / Delete Policy

No cascade that can remove Audit, Acceptance, Corrective, SLA breach or completed/missed Visit history. Master deactivation blocks future selection but does not erase historical references. Retention/archive is platform policy.

7. Security & Data Protection

Control	Design
Tenant/site isolation	Derived authenticated context on reads/actions/downloads.
IDOR	Non-leaking deny/not-found; no metadata leakage.
PII	Acceptance contact email/phone and file content Restricted; sensitive access controlled/audited.
Files	Private object reference + SHA-256 + scan status.
Secrets	No credentials/tokens in business tables.
Audit	Append-only; correlation links business transaction and notification.

8. Technical Decisions / Deferred Production Settings

Authentication technology, provider selection, retention duration, backup/RTO/RPO and observability are architecture/deployment settings. They do not alter the locked business semantics.

9. Development Readiness

Database Design v1.1 is internally consistent with RR-DD-001 v1.3 and ready for SQL DDL / EF Core mapping after technical baseline approval.

Performance Audit Revision Appendix

ส่วนนี้เป็น revision content ที่มีผลเหนือ technical note เดิมเฉพาะเมื่อมีข้อความขัดกัน โดยไม่เปลี่ยน Business Baseline.

10. Database Performance Corrections

10.1 Notification inbox/retry index correction

Previous shorthand that implied `notification_recipient.created_at` is removed. Canonical model: event time is `notification_event.created_at` ; recipient has `delivery_status` , `attempt_count` and `last_attempt_at` .

Candidate inbox access path starts from recipient scope (for example `recipient_user_id` + `delivery_status` + `notification_event_id`) then joins `NotificationEvent` for event-time ordering. Final key order/INCLUDE/filtered index is verified with the real SQL and execution plan; this document does not add a nonexistent timestamp field merely to satisfy an index.

10.2 Query-plan verification rule

Index Strategy remains a **candidate aligned to access patterns**, not a command to add every index blindly. Before Performance Test, capture generated SQL and actual execution-plan/Query Store evidence for Request List/Search, duplicate warning, SLA monitor, technician schedule, approval inbox, audit timeline and notification inbox/retry.

10.3 Worker claim / contention boundary

SLA/outbox workers require an atomic, idempotent database claim so parallel workers cannot process one logical row twice. Exact lock hints/isolation/batch/concurrency are Technical Implementation choices validated against SQL Server/Azure SQL contention evidence; they are not Business Rules.

11. Scaling / Deep Paging Decision

OFFSET/FETCH vs keyset/seek pagination is selected per real filter/sort pattern. Keyset pagination is **not mandatory**. Test representative deeper pages and switch only when measured cost justifies it without changing user-visible semantics.

PTP-01 - Repeatable Load Test Profile (Technical Test Design)

Item	Technical Test Design - not a new Business Requirement
Baseline concurrency	100 concurrent virtual users - existing baseline.
Dataset	100,000 Repair Request records per Site - existing baseline. Related WO/Visit/Audit/ Notification data must be representative enough to exercise joins and security scope.

Item	Technical Test Design - not a new Business Requirement
Security scope	Use at least 2 tenants and 2 Sites in the test fixture where practical; every request is authenticated. Cross-scope negative traffic is measured separately.
Ramp / duration	Default proposal: 5-minute ramp-up, 30-minute steady-state, 5-minute ramp-down, plus one warm-up run. This may be adjusted in a versioned Test Plan and is not a business requirement.
Filters / sorting / paging	Use representative filters, common sort, first/middle/deeper pages and the same server-side tenant/site scope as production code.
Evidence	Build/commit, environment, DB migration, dataset seed/version, script version, Average/p50/p95/p99, throughput, unexpected error count, API/DB latency, CPU/memory, DB waits/blocking/slow-query evidence and trace samples.